REAL ESTATE BUYER SEGMENTATION

Project Methodology & Final Results Report

Final solution: K-Means clustering, K = 7

Dataset: 2,000 buyer profiles linked to 7,305 sold-property records

Prepared from the project results/screenshots shared in this conversation. The underlying CSV files and notebook were not independently re-run or audited for this report.

Project objective: group real-estate buyers with similar purchasing behaviour and profile characteristics, then use the groups to inform targeted sales, marketing, and customer-engagement ideas.

1. Executive Summary

This project used unsupervised learning to segment real-estate buyers. The workflow included transaction-to-buyer aggregation, feature selection, categorical encoding, standardization, K-Means clustering, cluster profiling, and analyst-assigned segment names.

The final documented model uses **K=7** for 2,000 buyers. Cluster sizes range from 48 (2.4%) to 496 (24.8%). The largest group is Satisfied Home Buyers (Cluster 2); the smallest is High-Frequency Individual Buyers (Cluster 6).

The reported silhouette score for K=7 is **0.289**, the highest among the listed candidate values. This is a modest score: the profiles are useful for exploratory interpretation, but groups overlap and should not be treated as perfectly distinct.

Project item	Reported value
Buyer profiles	2,000 rows; 13 columns
Sold properties	7,305 records
Merged transaction dataset	7,305 × 25
Clustering input	2,000 × 6 encoded/scaled features
Algorithm / final K	K-Means / 7
Silhouette (K=7)	0.289
Outputs	Cluster assignment per buyer; seven segment profiles

2. Dataset and Data Quality

2.1 Data grain and aggregation

The transaction-level merged table contained 7,305 sold properties and 25 columns. Buyer-level aggregation produced one row per unique client (2,000 buyers), including properties purchased, total purchase value, average purchase price, average floor area, client type, gender, country, region, acquisition purpose, satisfaction, loan application, and referral channel.

Table	Reported shape/count	Role
Buyer/client source	2,000 × 12; cleaned 2,000 × 14	Client attributes
Sold-property data	7,305 rows	Property/transaction details
Merged table	7,305 × 25	Transactions joined to buyer data
Buyer-level profile	2,000 × 13	One record per buyer
Selected model features	2,000 × 6	K-Means input

2.2 Checks reported

Validation	Result
Duplicate listing IDs	0
Duplicate buyer-property pairs	0
Missing values in listed key fields	0
Unique buyers	2,000
Unique sold properties	7,305
Missing buyer profiles after merge	0

Age/DOB note: an initial date parsing error occurred because date strings used mixed separators/formats. A later notebook output reported 2,000 valid ages, no missing ages, age range 25–95, and mean age 55.9. DOBs were interpreted using Month-Day-Year; confirm this against the source convention before relying on age results.

3. Exploratory Analysis

3.1 Acquisition purpose

Purpose	Buyers	Share
Home	1,385	69.25%
Investment	615	30.75%
Total	2,000	100%

3.2 Loan applications

Purpose	No	Yes	Yes rate
Home	883	502	36.25%
Investment	381	234	38.05%

3.3 Satisfaction

Overall satisfaction scores were: 1 = 395, 2 = 379, 3 = 404, 4 = 417, and 5 = 405. Mean satisfaction was 3.02 for Home buyers and 3.04 for Investment buyers; median was 3 for both.

3.4 Geography

Buyer counts by country: USA 1,538 (76.9%), UK 95, Canada 85, Germany 56, France 53, Belgium 43, Mexico 40, Australia 39, Russia 36, and Denmark 15. California led regions with 633 buyers, followed by Nevada 143, Colorado 118, Arizona 108, and Oregon 96. These are dataset counts, not estimates of total market demand.

4. Methodology

4.1 Buyer-level features

Transaction records were aggregated to the buyer level. Six features were selected for clustering:

Feature	Definition
properties_purchased	Count of properties purchased by a buyer
avg_purchase_price	Mean purchase price across the buyer’s properties
avg_floor_area	Mean property floor area
satisfaction_score	Buyer rating (1–5)
acquisition_purpose	Home or Investment
client_type	Individual or Company

Client identifiers and descriptive fields were excluded from the model matrix.

4.2 Encoding and scaling

Acquisition purpose and client type were one-hot encoded. The reported encoded columns were acquisition_purpose_Investment and client_type_Individual, together with the four numeric variables, giving a 2,000 × 6 matrix. Standardization then produced column means approximately 0 and standard deviations approximately 1, preventing large-unit variables such as price and floor area from dominating distance calculations.

4.3 Selecting K

The elbow method was reviewed for K=1–10. Inertia declined as K increased, with a bend around K=4–5. Silhouette scores were also compared:

K	Silhouette
3	0.247

K	Silhouette
4	0.281
5	0.259
6	0.274
7	0.289
8	0.277

K=7 was selected because it had the highest silhouette score among the reported candidate K values. The value 0.289 indicates only modest separation.

4.4 Fit and profiling

K-Means was fitted to the standardized six-feature matrix. Labels 0–6 were attached to buyer profiles. The clusters were summarized using buyer counts, numerical averages, and categorical percentages. Segment names were assigned after reviewing the observed profiles.

5. Final K=7 Results

5.1 Segment size and assigned names

Cluster	Segment name	Buyers	Share
0	Premium Home Buyers	399	19.95%
1	Individual Investment Buyers	326	16.30%
2	Satisfied Home Buyers	496	24.80%
3	Low-Satisfaction Home Buyers	392	19.60%
4	Larger-Property Investors	237	11.85%
5	Corporate Property Buyers	102	5.10%
6	High-Frequency Individual Buyers	48	2.40%
Total		2,000	100%

5.2 Numeric profiles

Cl.	Avg properties	Avg total purchase value	Avg purchase price	Avg floor area (sq ft)	Avg satisfaction
0	3.44	1,478,320.34	431,199.10	1,416.35	2.81
1	3.56	1,055,098.88	296,773.69	985.59	2.90
2	3.70	1,186,392.92	320,824.55	1,065.44	4.21
3	3.56	1,072,649.55	301,399.41	1,002.73	1.71
4	3.47	1,420,361.42	410,221.86	1,343.78	3.17
5	3.61	1,240,325.73	345,480.71	1,145.10	3.05
6	7.38	2,393,135.81	325,918.58	1,077.65	3.54

Values reproduce the shared profile output. Currency/unit conventions were not consistently visible in the screenshots; confirm them from the source dataset. These are descriptive profile averages, not standardized-space K-Means centroids.

5.3 Categorical composition

Cluster	Home / Investment	Company / Individual	Profile signal
0	100% / 0%	0% / 100%	Home-focused individuals; highest mean price and floor area
1	0% / 100%	0% / 100%	Investment-focused individuals
2	100% / 0%	0% / 100%	Home-focused individuals; highest satisfaction
3	100% / 0%	0% / 100%	Home-focused individuals; lowest satisfaction
4	0% / 100%	0% / 100%	Investment-focused; relatively large properties
5	65.7% / 34.3%	100% / 0%	Company buyers; mixed purpose
6	64.6% / 35.4%	2.1% / 97.9%	Mostly individual; highest purchase frequency

Cluster 6 had 48 buyers; properties purchased averaged 7.38, with observed values from 6 to 13.

6. Interpretation and Possible Business Actions

Segment	Possible action (to validate)
Premium Home Buyers	Curate premium home inventory and high-touch property advisory; validate budget and service preferences.
Individual Investment Buyers	Share investment-focused listings and relevant property/return comparisons; learn which criteria influence decisions.
Satisfied Home Buyers	Explore referrals, reviews, and repeat-purchase opportunities; identify drivers of their satisfaction.
Low-Satisfaction Home Buyers	Conduct service-recovery follow-up; investigate property-fit or process pain points before additional offers.
Larger-Property Investors	Highlight larger-area investment options and compare by location, price, and intended use.
Corporate Property Buyers	Consider account-based outreach, portfolio discussions, and a dedicated corporate sales path.
High-Frequency Individual Buyers	Offer relationship/portfolio services; validate what motivates repeated purchases.

These are hypotheses suggested by descriptive profiles. No campaign experiment, causal analysis, profitability-by-segment, or response-lift evaluation was provided.

7. Evaluation, Limitations, and Next Steps

Evaluation

K=7 was the best-scoring option among the reported candidates (silhouette 0.289). It yields interpretable differences in satisfaction, acquisition purpose, client type, purchase frequency, price, and floor area. However, separation is modest.

Limitations

- Results depend on selected variables, encoding, scaling, and K-Means assumptions.
- K-Means may be sensitive to outliers and initialization; stability across seeds/resamples was not reported.
- Silhouette 0.289 implies overlap; segment labels are analyst interpretations, not verified customer classes.
- DOB parsing uses a date-format assumption; validate source conventions.
- Currency and exact price-field definitions should be documented.
- No external validation or downstream business-impact test was supplied.

Recommended next steps

1. Verify DOB conventions, currency, and price/floor-area definitions.
2. Test cluster stability across random seeds and samples; compare K=5, K=7, and alternative methods.
3. Inspect outliers and skew; consider transformations or robust scaling where justified.
4. Explore additional relevant variables if available (e.g., location, property type, recency, financing, channel), avoiding leakage/redundancy.
5. Build a dashboard with segment size, characteristics, geography, and a clear exploratory-model caveat.
6. Pilot segment-specific actions and measure conversion, satisfaction, repeat purchase, and revenue.

8. Conclusion

The project created a seven-segment profile of 2,000 real-estate buyers using K-Means on six encoded and standardized features. The segments highlight contrasts in satisfaction, acquisition purpose, client type, purchasing frequency, average price, and floor area.

The most notable profiles include the high-satisfaction Home segment (Cluster 2), low-satisfaction Home segment (Cluster 3), premium Home segment (Cluster 0), corporate buyers (Cluster 5), and high-frequency individuals (Cluster 6). Use these groups as a basis for further analysis and targeted engagement, subject to stability checks and business validation.

Source note: this report summarizes the figures and screenshots shared in the conversation. Raw CSVs and the notebook were not independently audited during PDF creation.